



BIG Blue Is Coming!

IBM strikes back with Cell—hailed by some to be the most important processor since the Pentium. How does it kick competing hindquarters?

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With all the hype that's surrounding Sony's upcoming PlayStation 3 (PS3), you've no doubt heard that its heart and soul will be the Sony-Toshiba-IBM Cell processor (the Cell Broadband Engine, to be precise)—a nine-core monstrosity that will ensure buttery-smooth frame rates even in the most crowded scenarios, and generally ensure a wholesome gaming experience for all. So what is it that makes the Cell such a formidable force? Naturally, it's the nine processor cores on the chip. But if it were as simple as that, this article would be a waste of space, wouldn't it? Before we go into its innards, however, we need to take a look at why the Cell exists in the first place.

Please Sir, I Want Some More

In the Old Days, it was office applications that drove processor speeds—after all, what other use did we have for PCs anyway? However, as processors got faster, it wasn't just for the thrill of faster word processing that we bought them—a vicious cycle had begun. The more powerful processors got, the more we wanted them to do for us. Games and multimedia became more demanding, and soon it was these applications that would have manufacturers sweating to take out the next ultra-fast processor.

To get these applications performing at their best, it all boiled down to just one thing—CPUs needed to execute more instructions per second. The easiest way of doing this, of course, was to kick up the CPU's clock frequency, which triggered the "MegaHertz (and later, the GigaHertz)